

and decreased operational performance.

Poor customer experience: Data plays a crucial role in delivering personalised experiences to passengers. When airlines lack sufficient customer data, they may struggle to understand passenger preferences, anticipate their needs, and provide tailored services. This can result in a generic and impersonal customer experience, leading to decreased customer satisfaction and loyalty.

Inaccurate decision-making: Data-driven decision-making is critical for airlines to stay competitive. Unavailability of accurate and up-to-date data can result in suboptimal route planning, pricing strategies, and marketing campaigns, leading to missed revenue opportunities and diminished profitability.

Ineffective risk management: Inadequate data can hinder airlines' ability to assess and mitigate risks effectively. Without comprehensive data on weather patterns, air traffic congestion, or potential disruptions, airlines may struggle to proactively manage and respond to operational challenges. This can result in flight delays, cancellations, and customer dissatisfaction.

Limited forecasting and planning: Data is essential for forecasting future trends, anticipating market changes, and planning for the long term. Without access to robust data, airlines may find it challenging to accurately predict passenger demand, adjust capacity, or respond to market fluctuations. This lack of foresight can lead to missed opportunities or inadequate preparations for changing market dynamics.

Challenges in getting accurate data

Airlines face several challenges when it comes to obtaining adequate data in a timely manner.

Data volume: Airlines receive huge volumes of data and sometimes their on-premises database systems cannot expand dynamically to a spike in data.

Data fragmentation: Airlines often have data scattered across various systems, departments, and third-party providers, making it challenging to consolidate and integrate data from multiple sources into a cohesive and unified data set. Incomplete or disconnected data can limit the ability to gain a comprehensive view of operations or customer insights.

Legacy systems and infrastructure: Many airlines still rely on legacy systems that were not designed to handle the vast amounts of data generated in today's digital age. Outdated infrastructure and technology may lack the scalability and processing power needed to capture, store, and analyse large volumes of data in real-time. Upgrading or replacing legacy systems can be costly and time-consuming.

Data quality and standardisation: Ensuring data quality and standardisation is crucial for accurate analysis and decision-making. Airlines may encounter issues with inconsistent data formats, missing or erroneous data, and data duplication. This can lead to unreliable insights and hinder the ability to derive meaningful conclusions from the data.

Data security and privacy: Airlines deal with sensitive data, including passenger information, payment details, and operational data. Ensuring data security and privacy is a significant concern. Compliance with regulations such as GDPR (general data protection regulation) and maintaining robust cyber security measures add complexity to data management. Striking a balance between data accessibility and protection is essential.

Data silos and interoperability: Different departments within an airline may operate in silos, each with its own set of data and systems. Lack of interoperability and data sharing between departments can impede the ability to gain holistic insights. Integrating data across different systems and fostering collaboration

among departments is necessary for a comprehensive view of operations.

Real-time data availability: Accessing and processing data in real-time can be challenging due to various factors such as system latency, data transmission delays, and reliance on batch processing. Ensuring the availability and timeliness of real-time data can be a complex task.

Data governance and data culture: Establishing robust data governance practices and fostering a data-driven culture within an airline is vital. This includes defining data ownership, establishing data management frameworks, and ensuring data literacy among employees. Without proper governance and a culture that values data-driven decision-making, it can be difficult to prioritise data initiatives and utilise data effectively.

Getting a budget for investing in IT systems is a major challenge, which is why modern databases are sometimes not employed. Fortunately, cloud-based database systems have almost zero upfront investments.

Advantages of cloud managed databases

Small and big organisations move to the cloud for database hosting due to the scalability, cost efficiency, high availability, global accessibility, security, simplified management, and integration capabilities offered by cloud providers. These advantages allow businesses to focus on their core competencies while leveraging the benefits of cloud-based database solutions.

Cloud databases offer several advantages, including scalability without upfront investments, cost efficiency through pay-as-you-go pricing, high availability and reliability, global accessibility for remote collaboration, robust security and compliance measures, simplified management and automation, and seamless integration with other cloud services. These benefits drive businesses